RIFLE MARKSMANSHIP & EMPLOYMENT

13th Legion — Training Reference

Adapted from TC 3-22.9 (Rifle and Carbine), Chapters 5–9 · May 2016 / C3 Nov 2019

1. EMPLOYMENT — THE SHOT PROCESS

The rifleman's primary role is to engage the enemy with well-aimed shots. The rate of fire is based on how fast you can **consistently acquire and engage with accuracy**, not how fast you can pull the trigger.

Every accurate shot requires two things:

1. **Properly point the weapon** (sight alignment + sight picture).
2. **Fire without disturbing the aim** (trigger control).

The Three-Phase Shot Process

Every shot — whether slow fire or rapid engagement — follows this sequence:

PHASE	ACTIONS
Pre-Shot	Position → Natural Point of Aim → Sight Alignment / Picture → Hold determination
Shot	Refine Aim → Breathing Control → Trigger Control
Post-Shot	Follow-through → Recoil management → Call the Shot → Evaluate target

> **KEY:** The shot process may be interrupted at any point before the sear disengages. If the situation changes — don't take the shot. One cognitive task at a time. The level of attention given to each element is proportional to the difficulty of the shot.

Four Functional Elements

These elements are interdependent. An accurate shot requires all four:

- **Stability** — Create and maintain a stable firing platform through the entire shot process.
- **Aim** — Orient the weapon, align sights, align on target, apply appropriate hold.
- **Control** — All conscious actions: trigger control, breathing, workspace management, fire/no-fire decisions.
- **Movement** — Maintain stability, aim, and control while moving laterally, forward, diagonally, or retrograde.

2. TARGET ACQUISITION

Before any shot process begins, you must detect, identify, and prioritize threats.

Detection: Scan → Acquire → Locate

- **Rapid scan** — Quick sweep for obvious threat activity. First method used.
- **Slow scan** — Deliberate observation using optics/aiming devices. Best in defense or during halts.
- **Horizontal scan** — Sweeping scan across key areas in restricted/urban terrain.
- **Vertical scan** — Up/down scan in urban or elevated terrain.
- **Detailed search** — Methodical review through optics of locations where you would position yourself if you were the threat.

> **BEST PRACTICES:** Scan with unaided eye first, then magnified optics. Don't search the same area as teammates — overlap sectors, don't duplicate. Think like the threat: search where it's most advantageous from their perspective.

Target Identification

Positively identify every target as **friend**, **foe**, or **noncombatant** before engaging. Use all available information, optics, and recognition features (markings, panels, beacons).

Target Prioritization

THREAT LEVEL	DEFINITION	ACTION
Most Dangerous	Has capability to defeat you AND is preparing to do so	Engage immediately
Dangerous	Has capability but is not prepared	Engage after most dangerous eliminated
Least	Cannot defeat you directly but can coordinate with	Engage last

Dangerous	higher threats

When multiple targets are at the same threat level:

- **Near** before far
- **Frontal** before flank
- **Stationary** before moving

3. STABILITY

Stability is built from four functions: **support**, **muscle relaxation**, **natural point of aim**, and **recoil management**.

Support

Support comes from your body (bones and muscles) and artificial support (sandbags, barricades, bipods, terrain). **The more support, the more stable the weapon.**

Body Mechanics

- **Firing hand:** Firm handshake grip on pistol grip — back strap high in the web between thumb and index finger. No gap between middle finger and trigger guard. Trigger finger indexed on lower receiver until ready to engage.
- **Non-firing hand:** As far forward on the handguard as comfortable. Thumb applies downward pressure on the handguard to control muzzle rise. Firm but relaxed grip.
- **Firing elbow:** Provides consistent grip and support; placement varies by position.
- **Non-firing elbow:** Positioned directly under the rifle when possible.
- **Buttstock:** Placed high in the shoulder pocket. Higher positions from the ground = higher buttstock in the shoulder. Side-to-side placement varies with equipment.
- **Stock weld:** Rest the *full weight* of your head on the stock. Head as upright as possible. **Bring the rifle up to your head, not your head down to the rifle.** Bony portion of the cheek on the stock is the starting point — adapt to your facial structure for repeatable placement.

Muscle Relaxation

- Strained or fatigued muscles destroy stability.
- More bone support = less muscle required = longer you can hold the position.
- Skeletal support does not fatigue. Muscle support does.

Natural Point of Aim (NPA)

NPA is where the barrel naturally orients when your muscles are relaxed and support is achieved.

- The closer NPA is to the target, the less muscle support required.
- More of your body on the ground = more stable = less mobile.
- If your wobble area is larger than the target, you need a steadier position.

> **NPA CHECK:** Assume your position. Get to the natural respiratory pause. Close your eyes, go through one breathing cycle, open your eyes. Where the sights are sitting is your NPA. If it's not on the target, adjust your *body position* (not your arms/grip). Repeat until NPA matches point of aim.

Recoil Management

Proper recoil management = the sights rapidly return to target after each shot = faster follow-up shots. This comes from a stable position, body weight behind the weapon, and correct buttstock placement. The shooter-gun angle (relationship between upper body and weapon direction) directly affects recoil control and changes between positions.

Carry Positions (Low to High Readiness)

POSITION	WHEN TO USE	KEY DETAILS
Hang	Hands needed, no threat present	Weapon slung, safety on. Do not use when weapon status is RED.
Safe Hang	No immediate threat, hands free	Slung, safety on, firing hand on pistol grip. Muzzle oriented at ground.
Collapsed Low Ready	Crowded/restrictive environments	Both hands on weapon, pulled in close. Minimizes flagging.
Low Ready	Contact possible, maximum observation	Buttstock in shoulder, muzzle 30–45° down toward sector. Head upright, looking over sights. Highest readiness + best field of view.

High Ready	Overhead sector / elevated muzzle needed	Buttstock in armpit, muzzle 45°+ up. Less effective than low ready — impedes field of view.
Ready (Up)	Contact imminent / engaging	Buttstock in shoulder, muzzle on threat or likely direction. Looking through sights. Finger off trigger until decision to engage.

Firing Positions (Most to Least Stable)

Lower positions = more ground contact = more stability. The situation and tactics determine which position to use.

Prone Supported

Most stable. Artificial support (sandbags, pack, berm) under the handguard. Legs spread, heels flat or firing-side leg bent to relieve stomach pressure. Firm handshake grip; non-firing hand on support to maximize weapon control. *The barrel must not touch the support* — only the handguard.

Prone Unsupported

Same body position; non-firing hand supports the handguard directly. Magazine may rest on the ground — this will NOT cause a malfunction.

Prone Roll-Over / Reverse Roll-Over

Roll-over: Shoot under low obstacles. Sights rotated to the side — bullet impact will be lower and toward the direction of rotation. Error increases with distance.

Reverse roll-over: Cover too low for traditional prone. Most effective way to get supported behind very low cover. Significant trajectory offset at distance.

Sitting Positions

Three variations, all providing good stability with body weight behind the weapon:

- **Open-leg** — preferred with combat equipment. Both elbows inside knees.
- **Crossed-ankle** — broadest base of support. Non-firing ankle over firing ankle.
- **Crossed-leg** — may cause increased pulse/wobble from restricted blood flow.

All: bend forward at waist, elbows on or inside legs (not on kneecaps), firm stock weld.

Kneeling Unsupported

Non-firing elbow directly under the rifle, placed in front of or behind the kneecap — **never directly on the kneecap** (it will roll). Non-firing leg bent ~90°, directly under the rifle. Lean slightly forward for recoil management.

Aggressive (stretch) kneeling: All weight on non-firing foot, thigh to calf, upper body leaning forward, non-firing triceps on knee, firing leg stretched behind. Highly effective for rapid fire and movement.

Kneeling Supported

Same as unsupported but non-firing hand/elbow pushed against artificial support. *Barrel must not touch the support.* Forward pressure for recoil management.

Squatting

Rapid engagement when obstacles block standard positions. Squat as low as possible, back of triceps on knees (no bone-on-bone), then return to standing. Magazine against opposite forearm can aid stability.

Standing Unsupported

Least stable. Use for close targets or when no time for a steadier position. Lean slightly forward for recoil management.

Standing Supported

Use structures/barricades. Non-firing hand holds handguard firmly against support. Forward pressure from rear leg and upper body. *Barrel must not contact the support* — only the handguard.

> **CRITICAL:** The barrel must NEVER be in direct contact with artificial support. This causes erratic shots. Only the handguard rests on support.

4. AIM

Aiming Process (Stationary Targets)

1. **Weapon orientation** — Point the weapon toward the detected threat (horizontal and vertical).
2. **Sight alignment** — Align the aiming device with your eye:
    - *Iron sights:* Tip of front sight post centered in the rear aperture. **Focus on the front sight post** — the target will be blurry. This is correct.
    - *Optics (CCO/red dot):* Red dot visible and centered, proper eye relief.
    - *Magnified optics:* Full centered field of view with no shadow.

3. **Sight picture** — Place the aligned sights on the target.
4. **Point of aim** — Center of visible mass (what you can see, not what you estimate the full target to be).

> **REMEMBER:** The human eye can only focus on one object at a time. Focus on the front sight post / reticle, NOT the target. Consistent sight alignment requires the full weight of your head on the stock — if head placement changes between shots, groups will suffer. A 1/1000-inch deviation at the weapon = up to 18 inches off at 300 meters.

Common Aiming Errors

- **Non-dominant eye:** Shooters with cross-dominant eyes (e.g., right-handed, left-eye dominant) should consider shooting from their dominant eye side.
- **Incorrect zero:** No amount of good aiming compensates for a bad zero.
- **Incorrect sight alignment:** Failing to focus on the front sight post / reticle.
- **Incorrect sight picture:** Moving targets, concealment, or unaccounted wind.
- **Improper range determination:** Results in incorrect hold at distances beyond zero range.

Range Determination Methods

Front Sight Post Method

- Target **wider** than front sight post → less than 300m → use battlesight zero (point of aim = point of impact).
- Target **narrower** than front sight post → greater than 300m → hold-off required.

Recognition Method

- **100m** — Facial features clearly distinguished.
- **200m** — Face detail lost, skin/equipment color still identifiable.
- **300m** — Clear body outline, face color accurate, other details blurred.
- **400m** — Body outline clear, all detail blurred.
- **500m** — Body shape tapers at extremities, head indistinct from shoulders.

100-Meter Unit of Measure

Visualize a football field (100m). Estimate how many fit between you and the target. Accuracy decreases when terrain between shooter and target isn't fully visible.

Holds for Range

Immediate range holds are based on your zero. With a 300m zero (or 200m), center-of-mass works point of aim/point of impact inside that range. Beyond the zero distance, hold above center mass. The amount of hold increases with range — practice at known distances to build mental references.

Moving Targets

Two methods:

- **Tracking:** Track the target with the sights, maintain lead, squeeze trigger while maintaining lead.
- **Trapping:** Aim at a point ahead of the target's path, hold steady, fire as the target moves into the sight picture.

Lead is estimated based on target speed and range. Oblique targets (moving diagonally) appear slower — reduce the lead proportionally to the angle off the gun-target line.

Wind Compensation

Wind Direction (Clock System)

- **Full-value** (maximum effect): 2–4 o'clock, 8–10 o'clock
- **Half-value:** 1, 5, 7, 11 o'clock
- **No-value:** 6, 12 o'clock (headwind/tailwind — negligible lateral effect)

Wind Speed Indicators

- 0–3 mph: Smoke drifts, barely felt.
- 3–5 mph: Felt lightly on the face.
- 5–8 mph: Leaves in constant movement.
- 8–12 mph: Raises dust and loose paper.
- 12–15 mph: Small trees sway.

> **WIND HOLD:** Aim *into* the wind. Wind from the right pushes bullets left — hold right. Focus on the wind at the **halfway to two-thirds point** between shooter and target — that's where the bullet has lost velocity and is most affected. When in doubt, aim center mass.

5. CONTROL

The shooter is the weapon's fire control system, ballistic computer, and stabilization platform. Control is everything you consciously manage before, during, and after the shot.

Trigger Control

The most important — and most difficult — skill to master. Trigger control and stability are inseparable.

- **Finger placement:** Place the finger where it naturally falls on the trigger after achieving proper grip. There is no single correct point — it varies by hand size. The goal is smooth, straight-to-the-rear pressure.
- **Trigger squeeze:** Smooth, consistent rearward pressure until the weapon fires. Regardless of speed, the squeeze is always smooth — never a jerk or snatch.
- **Trigger reset:** After the shot fires, maintain focus on sights. Release trigger pressure just enough to hear/feel the sear reset (metallic click). The trigger is now pre-staged for the next shot.

Breathing Control

Breathing causes unavoidable body movement. Firing on the **natural respiratory pause** (the bottom of a normal exhale) minimizes this movement. If the shot doesn't break before discomfort, resume breathing and start over.

> **NOTE:** Vertical dispersion during grouping is most likely caused by poor aiming or trigger control — not breathing.

Workspace Management

The workspace is a 12–18 inch sphere centered on your chin, about 12 inches in front. This is where all weapon manipulations happen — magazine changes, charging handle, bolt catch, selector lever, malfunction clearance. Working in the workspace keeps your eyes oriented downrange toward the threat.

Calling the Shot

Know exactly where the sights were when the weapon discharged. Call the shot in clock direction and distance from the desired point of aim. This is how you diagnose errors and improve. **You are responsible for the point of impact of every round you fire** — including what's behind the target.

Follow-Through

After the shot fires:

1. **Recoil management** — Let the BCG cycle fully and return to battery.
2. **Recoil recovery** — Return to pre-shot position, reacquire sight picture.
3. **Trigger/sear reset** — Release trigger just enough for the sear to reset.
4. **Sight picture adjustment** — Get sights back on the aiming point.
5. **Engagement assessment:**
    - *Subsequent engagement* — Target needs more rounds. Resume shot process.
    - *Supplemental engagement* — Target down, another target needs servicing.
    - *Sector check* — All threats serviced. Scan your sector for additional threats.

Rate of Fire

TYPE	RATE	WHEN
Slow semi	~12–15 rpm	Deliberate engagements, defensive positions. Maximum focus on fundamentals.
Rapid semi	~45 rpm	Multiple targets, no overmatch. Only after slow semi is mastered.

6. MALFUNCTIONS

> **PRIORITY:** When a malfunction occurs, the priority is to defeat the target as quickly as possible. If you have a secondary weapon and the threat is close — transition immediately, then clear the malfunction from cover.

Immediate Action (SPORTS)

Used when: trigger squeezed → audible "click" → no bang.

1. **Tap** the bottom of the magazine firmly.
2. **Rack** — pull charging handle fully rearward and release.
3. **Reassess** — continue the shot process.

If the same symptom recurs: remove magazine, insert a fresh loaded magazine, repeat above.

Remedial Action

Required when: immediate action fails after two attempts, trigger feels like "mush," or trigger cannot be squeezed.

1. Look into the ejection port to identify the malfunction type.
2. Remove magazine.
3. Attempt to clear the weapon.
4. Visually inspect chamber, bolt face, and charging handle.
5. Apply the specific fix:

MALFUNCTION	SYMPTOM	CORRECTIVE ACTION
Stove pipe	Casing lodged sideways in ejection port	Grasp case and remove, cycle weapon. If stuck, pull charging handle rearward while holding case.
Double feed	Round chambered + second round being fed	Remove magazine → clear weapon → confirm chamber clear → seat new magazine → chamber round.
Bolt override	Bolt riding on top of a cartridge	Remove magazine → pull charging handle fully rearward → strike charging handle forward. If it fails, pull CH back again, hold bolt to rear with finger/tool, sharply send CH forward.

Rules for Clearing Malfunctions

- Keep treating the weapon as loaded (Rule 1).
- Keep muzzle oriented appropriately — don't flag friendlies (Rule 2).
- Finger off the trigger, straight along the lower receiver (Rule 3).
- **Do NOT attempt to put the weapon on SAFE** — wastes time and may damage the weapon when out of battery.
- **Treat the symptom.** Don't analyze — execute the most probable corrective drill.
- **Maintain focus on the threat**, not the weapon. Look downrange, not at the malfunction.
- **Look last.** Only visually inspect the weapon after the corrective action is executed and the threat is eliminated.

7. MOVEMENT

The most complex form of shooting: moving while engaging threats. Sight alignment and trigger control are at their most critical during movement.

General Principles

- **Roll the foot heel-to-toe** to provide the smoothest, most stable platform.
- **Bend the knees** — use them as shock absorbers to isolate the upper body from lower body movement.
- Keep hips as stationary as possible. **Use the upper body as a turret**, twisting at the waist.
- Walk close to your natural gait — don't overstep or cross feet.
- Lean forward aggressively to keep mass behind the weapon for recoil management.
- Trigger control while moving is based on the wobble area — **shoot when the sights are most stable**, not based on foot position.

Movement Types

Forward

Movement toward the threat. Most basic form. Roll heel-to-toe, keep feet almost in-line (reduces visible "bouncing" of sight picture), muzzle oriented downrange.

Retrograde

Moving rearward while keeping the weapon oriented on the threat. Reverse the gait: **toe-to-heel**. Take only one or two steps at a time. Bend knees, lean forward at the waist. Maintain situational awareness for obstacles behind you.

Lateral

Moving left or right while maintaining weapon orientation on the threat. Heel-to-toe, don't cross feet (compromises balance). For extreme lateral angles (90°+), drop the non-firing shoulder and roll the upper body to gain additional offset.

Turning

Engaging widely dispersed targets on the flanks. Sequence: **Look first** (turn head) → **Weapon follows the eyes** → **Body follows the weapon**. Maintain an exaggerated low-alert carry throughout the turn. Muzzle stays down until the body is fully oriented on the threat.